

Notes: Brogaard, Nguyen, Putnins, and Wu (RFS, 2022)

What Moves Stock Prices? The Roles of News, Noise, and Information

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1 Big picture

- **Question.** What fraction of stock-return variation comes from information and what fraction comes from noise? Within information, how much is market-wide information, private firm-specific information revealed through trading, and public firm-specific information?
- **Core contribution.** The paper builds a return variance decomposition that separates transitory price movements from permanent price movements and further separates permanent movements by information source.
- **Baseline result.** For U.S. common stocks, the average return-variance decomposition is approximately 8.6% market-wide information, 24.0% private firm-specific information, 36.7% public firm-specific information, and 30.7% noise.
- **Time-series result.** Since the mid-1990s, noise has fallen sharply and firm-specific information has increased. The decline in noise is particularly visible after 1997.
- **Market-efficiency implication.** The traditional R^2 measure can be misleading because it treats all idiosyncratic variation as firm-specific information. If idiosyncratic noise falls, R^2 can rise even though market efficiency improves.
- **Cross-sectional result.** Large stocks have less noise and more private firm-specific information than small stocks. HiTech and Healthcare stocks show high private firm-specific information shares.
- **Validation tests.** The decomposition behaves as expected around plausibly exogenous shocks: tick size reductions reduce noise and increase private information; analyst-coverage losses reduce public information and increase noise; discretionary mutual fund trading increases private information and reduces noise.
- **Economic takeaway.** Stock prices move because of both information and noise. Measuring market efficiency requires distinguishing permanent information-driven price changes from temporary noninformational deviations.

2 Incremental contribution

2.1 Contribution relative to classic return-decomposition approaches

The paper advances two classic literatures. First, the Campbell-Shiller-Campbell return-decomposition literature separates returns into cash-flow and discount-rate news. Second, the Morck-Yeung-Yu style literature uses market-model R^2 to interpret the market-wide versus firm-specific components of stock returns. Brogaard, Nguyen, Putnins, and Wu add a third dimension: they explicitly separate information from noise.

- Prior R^2 decompositions cannot tell whether idiosyncratic return variation is firm-specific information or firm-specific noise.
- Prior cash-flow/discount-rate decompositions typically abstract from transitory noninformational deviations from efficient price.
- This paper builds a model in which a return can move because efficient price changes or because actual price temporarily deviates from efficient price.

Interpretation: The incremental contribution is not simply a new empirical statistic. It changes the interpretation of common facts. For example, an increase in R^2 can reflect less idiosyncratic noise rather than less firm-specific information.

2.2 Contribution relative to market microstructure

The paper uses tools familiar in market microstructure, especially vector autoregressions, impulse responses, order-flow shocks, and permanent-transitory decompositions. It applies these tools at a large-sample stock-year

level to answer broader asset-pricing questions.

- It links order-flow innovations to private firm-specific information.
- It links return residual innovations not explained by market returns or order flow to public firm-specific information.
- It measures transitory deviations as noise, including price pressure, overreaction, underreaction, liquidity effects, and microstructure frictions.

Interpretation: The paper turns microstructure VAR methods into a scalable variance-decomposition tool that can be applied to thousands of stocks over many decades.

2.3 Contribution to empirical market-efficiency debates

The paper provides direct evidence that U.S. stock prices became less noisy and more firm-specific-information-driven over time. This is important because earlier work often interpreted rising R^2 or declining idiosyncratic volatility differently.

- The decomposition suggests that market efficiency improved after the mid-1990s because noise declined.
- It reconciles rising R^2 with increasing informativeness by showing that noise is an omitted component in traditional interpretations.
- It provides a framework for evaluating the effects of market structure changes, analyst coverage, institutional trading, and algorithmic trading.

Interpretation: The paper’s incremental contribution is to make “price informativeness” empirically multidimensional: prices can become more informative because they incorporate more public information, more private information, or less noise.

3 Data and measurement

3.1 Stock sample

- The main sample consists of common stocks listed on the NYSE, AMEX, and NASDAQ.
- The main variance-decomposition sample runs from 1960 to 2015.
- The extended time-series analysis reaches back to 1926 where the required data are available.
- The paper uses CRSP daily returns, prices, volumes, shares outstanding, and market capitalization.
- It uses Compustat for firm accounting variables and IBES/13F-related data for some validation exercises.

Interpretation: The stock-year is the main unit of analysis. The model is estimated separately for each stock-year using daily data, and then variance shares are averaged across stocks.

3.2 Core reduced-form variables

For each stock i on day t , the VAR uses:

- market return, $r_{m,t}$;
- signed dollar volume, $x_{i,t}$, which proxies for order-flow pressure or net buying;
- stock return, $r_{i,t}$.

The baseline VAR has five lags:

$$y_{i,t} = A_1 y_{i,t-1} + A_2 y_{i,t-2} + \cdots + A_5 y_{i,t-5} + \varepsilon_{i,t}, \quad (1)$$

where

$$y_{i,t} = \begin{bmatrix} r_{m,t} \\ x_{i,t} \\ r_{i,t} \end{bmatrix}. \quad (2)$$

Interpretation: The three innovations are interpreted as market-wide shocks, trading/order-flow shocks, and residual firm-specific return shocks.

3.3 Economic interpretation of the three shocks

- **Market-return shocks** identify market-wide information.
- **Order-flow shocks** identify private firm-specific information revealed through trading.
- **Stock-return residual shocks** identify public firm-specific information, because they move the stock but are not captured by market returns or signed order flow.

Interpretation: The interpretation depends on the idea that informed private trading leaves an order-flow footprint, while public news can move returns without necessarily appearing first in signed order flow.

3.4 Permanent-transitory logic

For each shock, the VAR impulse response traces the cumulative stock-price response forward in time. The long-run stabilized price response is the permanent component:

$$\theta_j = \lim_{h \rightarrow \infty} E \left[\sum_{s=0}^h r_{i,t+s} \mid \varepsilon_{j,t} = 1 \right], \quad (3)$$

where j indexes the market, order-flow, or residual stock-return shock.

- The permanent component is interpreted as information.
- The difference between the actual immediate return and the permanent response is interpreted as noise.
- Noise can be positive or negative: overreaction and underreaction both generate transitory components.

Interpretation: This is the Beveridge-Nelson decomposition applied to stock prices. It uses long-run impulse responses to distinguish efficient-price innovations from temporary price pressure.

3.5 Variance shares

Let the permanent information components be market-wide, private firm-specific, and public firm-specific. Let the transitory component be noise. The variance shares are:

$$\text{MktInfoShare}_{i,t} = \frac{\text{Var}(\text{market-wide information})}{\text{Var}(\text{return})}, \quad (4)$$

$$\text{PrivateInfoShare}_{i,t} = \frac{\text{Var}(\text{private firm-specific information})}{\text{Var}(\text{return})}, \quad (5)$$

$$\text{PublicInfoShare}_{i,t} = \frac{\text{Var}(\text{public firm-specific information})}{\text{Var}(\text{return})}, \quad (6)$$

$$\text{NoiseShare}_{i,t} = \frac{\text{Var}(\text{noise})}{\text{Var}(\text{return})}. \quad (7)$$

Interpretation: These four components sum to approximately 100% of return variance in the baseline model. This is the central empirical object in the paper.

4 Empirical specifications

4.1 Panel regressions for determinants of variance shares

The paper estimates panel regressions of the form:

$$Share_{i,t} = \alpha + \gamma_1 D_t^{POST} + \sum_j \gamma_j D_{j,i}^{INDUSTRY} + \sum_k \gamma_k X_{k,i,t} + \sum_m \gamma_m Z_{m,i,t} + \varepsilon_{i,t}, \quad (8)$$

where $Share_{i,t}$ is one of the variance-component shares.

- D_t^{POST} is an indicator for the post-1997 period.
- Industry dummies capture systematic differences across sectors.
- $X_{k,i,t}$ includes baseline stock characteristics such as log price, log market capitalization, variance ratio, and delay.
- $Z_{m,i,t}$ includes extended controls such as institutional ownership, hard-to-value measures, illiquidity, bid-ask spreads, and leverage.

Interpretation: These regressions validate whether the decomposition behaves in ways that are economically sensible across stock characteristics.

4.2 Tick-size pilot regressions

To validate the noise component, the paper studies the 1997 tick-size pilot and estimates interactions of the post-pilot period with pilot-group indicators:

$$Share_{i,t} = \alpha + \beta D_t^{POST} + \sum_q \delta_q (D_t^{POST} \times Q_{q,i}) + \gamma X_{i,t} + \varepsilon_{i,t}. \quad (9)$$

Interpretation: If smaller tick sizes improve liquidity and reduce price discreteness, the model predicts less noise and more information in treated stocks.

4.3 Analyst-coverage shock regressions

The paper uses broker mergers and closures as exogenous shocks to analyst coverage:

$$Share_{i,t} = \gamma_i + \delta_t + \beta CoverageShock_{i,t} + \varepsilon_{i,t}. \quad (10)$$

Interpretation: If analyst coverage supplies public information, a coverage shock should reduce public firm-specific information and may increase noise.

4.4 Mutual fund trading shock regressions

The paper uses mutual fund trading shocks decomposed into discretionary and expected trading:

$$Share_{i,t} = \gamma_i + \delta_t + \beta_1 DiscretionaryTradeShock_{i,t} + \beta_2 ExpectedTradeShock_{i,t} + \varepsilon_{i,t}. \quad (11)$$

Interpretation: Discretionary mutual fund trading should contain information and therefore increase private firm-specific information. Expected fire-sale trading is more mechanical and should have weaker information effects.

5 Identification and validation design

5.1 What identifies the decomposition

- The decomposition uses dynamic responses to shocks rather than only contemporaneous return correlations.
- Permanent components are identified by long-run impulse responses.
- Noise is identified by the temporary part of price changes that reverses or corrects over time.
- Economic labels are assigned using the source of the shock: market returns, signed volume, or stock-return residuals.

Interpretation: The model does not require observing every piece of news. It infers information from permanent price effects and noise from transitory price effects.

5.2 Why the validation tests are important

The decomposition could be mechanically estimated even if the labels were wrong. The paper therefore checks whether the estimated components respond sensibly to external shocks.

- Tick-size reductions should reduce noise.
- Analyst-coverage losses should reduce public firm-specific information.
- Discretionary mutual fund trading should increase private firm-specific information.
- Nondiscretionary mutual fund trading should have smaller effects on information.

Interpretation: These tests turn the decomposition from a statistical exercise into an economically interpretable measurement tool.

5.3 Limitations

- Signed order flow is an imperfect proxy for private information.
- Public news can be traded on quickly, so the separation between private and public information is not perfect.
- The VAR assumes a stable local dynamic structure within each stock-year.
- Variance shares are model-based, so interpretation depends on the impulse-response structure.
- The approach is strongest for broad comparisons and trends rather than for identifying every individual news event.

Interpretation: The paper is careful to present the decomposition as a measurement framework, not a direct observation of all information and noise.

6 Tables and figures

The visuals below are directly cropped from the paper. Nearby visuals are placed on the same row where possible. Single visuals use width $0.6\backslash\text{linewidth}$.

6.1 Figure 1 and Figure 2: decomposition intuition

Read:

- Figure 1 illustrates how Beveridge-Nelson impulse responses separate permanent information from transitory noise.
- A trading shock can generate overreaction: price initially moves too far and later reverts toward the efficient-price response.
- A nontrade news shock can generate underreaction: price initially moves too little and then drifts toward the long-run response.
- Figure 2 summarizes the baseline stock-return decomposition: noise, market-wide information, private firm-specific information, and public firm-specific information.

Economic message: The model's key idea is simple: the long-run response is information; the temporary gap between actual and efficient price is noise.

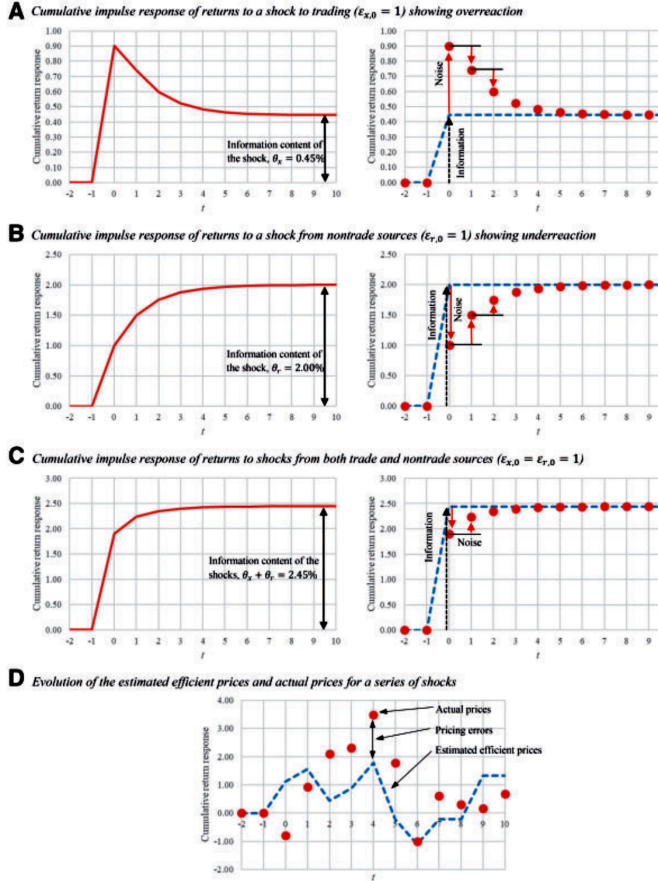


Figure 1: Beveridge-Nelson decomposition intuition

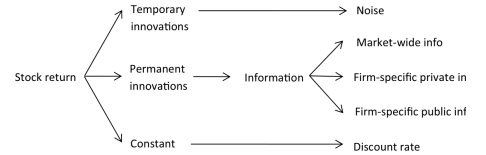


Figure 2

Figure 2: baseline return components

6.2 Table 1 and Table 2: VAR estimates and baseline variance shares

Read:

- Table 1 reports average VAR coefficients and correlations used to estimate the dynamic decomposition.
- Table 2 reports the headline baseline shares: 8.62% market-wide information, 24.00% private firm-specific information, 36.67% public firm-specific information, and 30.71% noise.
- Noise falls from 35.42% in 1960–1996 to 25.78% in 1997–2015.
- Public firm-specific information rises from 34.86% to 38.58%; private firm-specific information rises from 22.61% to 25.46%.
- Large stocks have much less noise than small stocks: 16.92% in Q4 versus 35.56% in Q1.

Economic message: The baseline results imply that roughly two-thirds of stock-return variance reflects information, while about one-third reflects temporary noise.

Table 1 VAR coefficient estimates		Coefficient average (% negative significant coefficients, % positive significant coefficients)					Correlation
Dependent variable	Independent variable	<i>l</i> =1	<i>l</i> =2	<i>l</i> =3	<i>l</i> =4	<i>l</i> =5	
<i>A. Market return equation</i>							
$r_{m,t}$	$r_{m,t-l}$	0.104 (2.09%, 37.93%)	-0.032 (8.13%, 0.34%)	0.018 (1.69%, 3.48%)	-0.005 (5.95%, 1.81%)	-0.010 (2.14%, 3.09%)	
	$r_{i,t-l}$	0.009 (2.36%, 2.83%)	0.024 (2.51%, 2.54%)	0.033 (2.41%, 2.46%)	0.081 (2.42%, 2.63%)	-0.034 (2.40%, 2.44%)	0.134 (0.57%, 42.81%)
	$r_{j,t-l}$	-0.002 (3.76%, 3.07%)	-0.001 (3.26%, 3.19%)	0.003 (3.04%, 3.14%)	0.003 (2.90%, 3.15%)	0.001 (3.22%, 2.83%)	0.329 (0.39%, 61.85%)
<i>B. Signed dollar volume equation</i>							
$s_{i,t}$	$r_{m,t-l}$	-0.650 (2.27%, 10.18%)	0.013 (2.42%, 2.91%)	0.012 (1.75%, 3.59%)	-0.529 (2.05%, 2.85%)	-0.785 (1.97%, 2.93%)	0.134 (0.57%, 42.81%)
	$r_{i,t-l}$	0.025 (8.69%, 14.74%)	-0.016 (9.21%, 5.75%)	0.003 (5.45%, 6.14%)	-0.008 (5.76%, 4.52%)	0.004 (4.26%, 5.15%)	0.615 (0.05%, 99.42%)
	$r_{j,t-l}$	1.736 (9.55%, 8.45%)	0.631 (4.66%, 8.35%)	0.196 (3.60%, 3.32%)	-0.002 (3.01%, 3.50%)	0.484 (2.76%, 3.19%)	0.625 (0.05%, 99.42%)
<i>C. Stock return equation</i>							
$r_{i,t}$	$r_{m,t-l}$	0.247 (2.39%, 21.77%)	0.045 (2.93%, 4.78%)	0.089 (1.71%, 6.02%)	0.052 (2.40%, 4.37%)	0.050 (2.36%, 4.56%)	0.229 (0.39%, 61.85%)
	$r_{i,t-l}$	0.072 (2.50%, 11.37%)	-0.077 (3.06%, 3.97%)	-0.059 (2.68%, 3.10%)	-0.150 (2.80%, 2.66%)	0.615 (2.65%, 2.51%)	0.615 (0.05%, 99.42%)
	$r_{j,t-l}$	-0.112 (31.77%, 5.47%)	-0.060 (16.40%, 2.61%)	-0.030 (8.88%, 2.68%)	-0.022 (6.27%, 2.58%)	-0.007 (4.27%, 3.07%)	

This table reports the mean coefficient estimates and the mean correlations between residuals for the baseline VAR model used to perform the variance decomposition. The VAR model is estimated separately for each stock in each year using daily observations. For the purpose of this table, each of the model coefficients is averaged across stocks and years and reported in the table. Below each coefficient average, in parentheses, we report the percentage of negative statistically significant (at 5%) coefficients (first number in the parentheses) and the percentage of positive statistically significant (at 5%) coefficients (second number in the parentheses). The correlation column is computed similarly, but rather than reporting coefficients it reports the correlations of the residuals for pairs of variables in the VAR. The variables used in the VAR are daily market returns in basis points ($r_{m,t}$), daily signed dollar volume in 5 thousands ($s_{i,t}$), and daily stock returns in basis points ($r_{i,t}$). The columns $l=1$ to $l=5$ correspond to lags of the independent variables. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ from 1960 to 2015 (average of 4,362 stocks per year for a total of 22,025 stocks).

Table 1: reduced-form VAR estimates

Table 2
Stock-return variance components in the baseline model

	<i>MktInfo</i> Share(%) [99% CI] or (<i>t</i> -statistic)	<i>PrivateInfo</i> Share(%) [99% CI] or (<i>t</i> -statistic)	<i>PublicInfo</i> Share(%) [99% CI] or (<i>t</i> -statistic)	<i>Noise</i> Share(%) [99% CI] or (<i>t</i> -statistic)
<i>A. Full sample</i>				
	8.62 [8.56–8.67]	24.00 [23.92–24.08]	36.67 [36.59–36.76]	30.71 [30.61–30.81]
<i>B. Subperiods</i>				
1960–1996	7.11 [7.04–7.19]	22.61 [22.50–22.72]	34.86 [34.75–34.97]	35.42 [35.28–35.56]
1997–2015	10.19 [10.10–10.28]	25.46 [25.33–25.58]	38.58 [38.45–38.71]	25.78 [25.65–25.91]
Difference (Post-Pre 1997)	3.07 (1.84)*	2.85 (1.83)*	3.72 (3.74)***	-9.64 (-4.85)***
<i>C. Quartiles by size (market capitalization)</i>				
Q1 = low	5.58 [5.51–5.66]	21.85 [21.70–22.00]	37.01 [36.85–37.16]	35.56 [35.37–35.76]
Q2	8.84 [8.73–8.95]	25.04 [24.87–25.21]	37.43 [37.24–37.62]	28.69 [28.48–28.90]
Q3	14.21 [14.07–14.36]	27.90 [27.72–28.08]	36.80 [36.62–36.99]	21.08 [20.91–21.25]
Q4 = high	21.09 [20.92–21.26]	30.33 [30.14–30.51]	31.67 [31.507–31.84]	16.92 [16.79–17.04]
Difference (Q1–Q4)	-15.51 (-14.82)***	-8.48 (-6.15)***	5.34 (5.79)***	18.64 (11.08)***
<i>D. Industry groups</i>				
Consumer	7.98 [7.85–8.10]	23.72 [23.53–23.90]	36.37 [36.18–36.56]	31.93 [31.71–32.16]
Healthcare	7.67 [7.49–7.85]	27.01 [26.71–27.31]	38.01 [37.70–38.33]	27.30 [27.00–27.64]
HiTech	9.50 [9.36–9.63]	25.26 [25.07–25.44]	37.12 [36.92–37.31]	28.13 [27.90–28.35]
Manufact	10.03 [9.89–10.17]	24.28 [24.11–24.45]	35.29 [35.12–35.47]	30.40 [30.18–30.61]
Other	7.75 [7.65–7.86]	21.89 [21.74–22.04]	36.80 [36.65–36.96]	33.56 [33.37–33.75]

This table reports the mean variance shares (expressed as percentages of variance) for the period from 1960

Table 2: baseline variance components

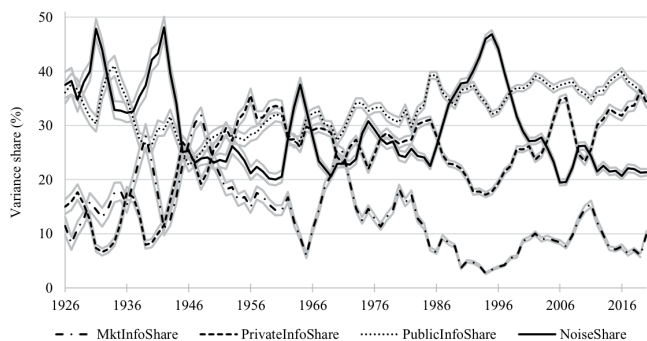
6.3 Figures 3 and 4: time-series trends and the R^2 contrast

Read:

- Figure 3 shows long-run trends in the four variance components.
- Noise rises into the early 1990s and then declines sharply after the mid-1990s.
- Firm-specific information becomes a larger share of return variance over time.
- Figure 4 shows that market-model R^2 falls through much of the earlier period and then rises sharply after the mid-1990s.
- The comparison reveals why R^2 can be misleading: a higher R^2 can reflect less idiosyncratic noise, not less firm-specific information.

Economic message: The decomposition resolves a key ambiguity in traditional measures of market efficiency.

A Extended time period 1926–2020



B Subperiod 1960–2015

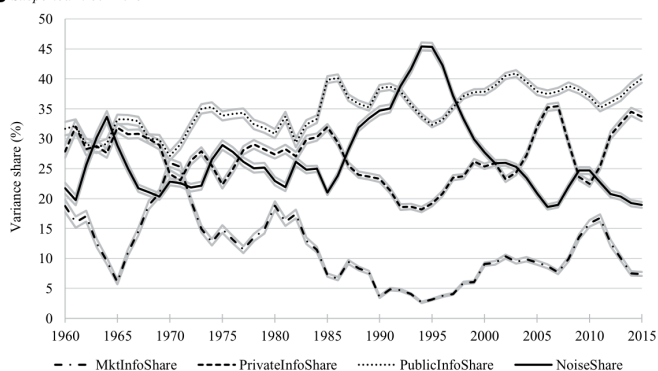


Figure 3: variance components through time

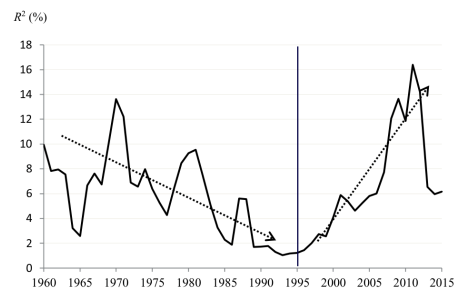


Figure 4: R^2 through time

6.4 Figures 5 and 6: cross-sectional trends by size and industry

Read:

- Figure 5 shows that all size groups exhibit declining noise after the mid-1990s.
- Smaller stocks have consistently higher noise and lower market-wide information shares.
- Figure 6 shows similar long-run patterns across industry groups.
- HiTech and Healthcare stocks have relatively high private firm-specific information shares.

Economic message: The trends are not driven only by one size group or one industry. The decline in noise and rise in firm-specific information are broad market phenomena.

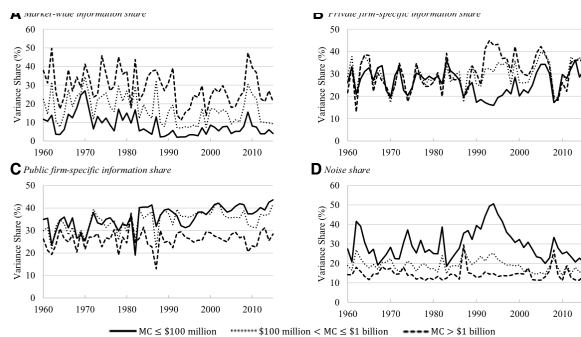


Figure 5: variance components by size group

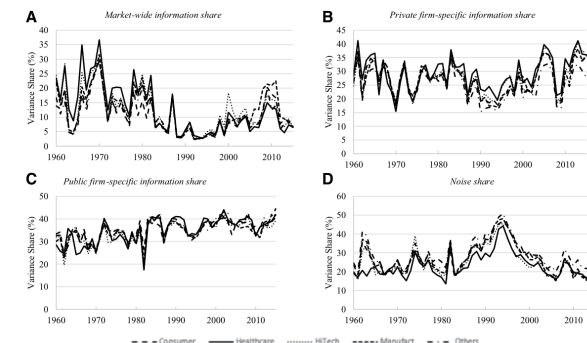


Figure 6

Figure 6: variance components by industry group

6.5 Table 3: determinants of variance shares

Read:

- Panel A shows that after 1997 stocks have lower noise shares and higher public firm-specific information shares.
- Larger stocks have lower noise shares and higher private firm-specific information shares.
- Higher delay is associated with lower market information and more public firm-specific information and noise.
- Panel B adds institutional ownership, hard-to-value measures, illiquidity, bid-ask spread, and leverage.
- Institutional ownership is positively related to private information and negatively related to noise.

Economic message: The determinants have sensible signs: stocks with better trading and more institutional monitoring tend to have more information and less noise.

Table 3
Determinants of stock-return variance shares

Variable	MktInfo Share _{i,t}	PrivateInfo Share _{i,t}	PublicInfo Share _{i,t}	Noise Share _{i,t}	(1 - R ²)
<i>A. Basic stock characteristics as determinants</i>					
Intercept	19.38 (7.03)***	12.51 (6.49)***	31.66 (18.80)***	36.45 (16.17)***	78.59 (34.12)***
D_t^{POST}	0.97 (1.03)	-1.37 (-1.34)	4.29 (5.60)***	-3.88 (-4.01)***	-2.30 (-3.34)***
$\ln P_{i,t}$	0.01 (0.02)	-1.02 (-1.95)*	0.21 (0.68)	0.80 (2.06)**	0.57 (1.97)**
$\ln MC_{i,t}$	0.60 (2.01)**	2.52 (8.92)***	-1.02 (-4.78)***	-2.09 (-7.35)***	-1.38 (-6.15)***
$D_t^{Consumer}$	-0.47 (-1.63)	4.40 (12.70)***	0.21 (0.53)	-4.13 (-9.10)***	1.36 (5.82)***
$D_t^{Healthcare}$	-1.07 (-2.15)**	6.64 (14.73)***	0.72 (1.39)	-6.29 (-13.43)***	2.99 (6.25)***
D_t^{HiTech}	0.58 (0.91)	4.77 (7.42)***	-0.07 (-0.16)	-5.27 (-10.58)***	1.80 (4.58)***
$D_t^{Manufact}$	0.99 (1.59)	2.75 (7.43)***	-0.16 (-0.31)	-3.58 (-7.93)***	-0.29 (-0.72)
$VarianceRatio_{i,t}$	-0.05 (-0.42)	0.59 (3.72)***	0.84 (7.17)***	-1.39 (-5.91)***	0.44 (2.54)**
$Delay_{i,t}$	-21.24 (-8.97)***	2.46 (1.45)	6.91 (5.93)***	11.87 (8.58)***	29.43 (10.56)***
Observations	137,903	137,903	137,903	137,903	137,903

Table 3A: baseline determinants

Table 3
(Continued)

Variable	MktInfo Share _{i,t}	PrivateInfo Share _{i,t}	PublicInfo Share _{i,t}	Noise Share _{i,t}	(1 - R ²)
<i>B. Extended set of stock characteristics as determinants</i>					
Intercept	17.32 (5.97)***	13.92 (5.28)***	36.25 (17.51)***	32.52 (12.46)***	80.17 (27.47)***
D_t^{POST}	1.78 (2.63)***	-2.59 (-2.36)**	3.26 (5.17)***	-2.44 (-2.18)**	-3.07 (-4.60)***
$\ln P_{i,t}$	-0.73 (-1.32)	-1.39 (-2.15)**	0.033 (0.11)	2.09 (6.28)***	0.71 (1.95)*
$\ln MC_{i,t}$	1.07 (5.91)***	2.27 (8.95)***	-1.39 (-10.16)***	-1.95 (-8.69)***	-1.49 (-6.68)***
$D_t^{Consumer}$	-1.10 (-2.88)***	4.97 (11.85)***	-0.10 (-0.23)	-3.77 (-7.34)***	2.36 (7.05)***
$D_t^{Healthcare}$	-1.62 (-2.51)**	6.82 (14.02)***	0.64 (1.08)	-5.83 (-10.54)***	3.78 (6.31)***
D_t^{HiTech}	0.29 (0.38)	5.23 (7.14)***	-0.43 (-0.83)	-5.08 (-9.09)***	2.37 (4.60)***
$D_t^{Manufact}$	0.75 (1.23)	3.19 (7.41)***	-0.26 (-0.51)	-3.67 (-6.67)***	0.19 (0.39)
$VarianceRatio_{i,t}$	-0.094 (-0.68)	0.52 (3.17)***	0.84 (7.00)***	-1.26 (-6.14)***	0.52 (2.46)**
$Delay_{i,t}$	-21.28 (-8.60)***	5.61 (2.78)***	8.60 (7.14)***	7.07 (4.22)***	31.28 (9.99)***
$InstoHold_{i,t}$	0.014 (0.96)	0.040 (2.32)**	-0.013 (-1.37)	-0.041 (-4.78)***	-0.04 (-2.56)**
$HardToValue_{i,t}$	-0.30 (-1.03)	1.88 (5.36)***	0.55 (2.38)**	-2.13 (-6.83)***	1.27 (4.58)***
$ILLIQ_{i,t}$	0.003 (1.88)*	-0.014 (-2.84)***	-0.015 (-2.24)**	0.026 (2.56)**	-0.012 (-2.67)***
$BidAskSpread_{i,t}$	0.013 (1.81)*	-0.068 (-3.45)***	-0.052 (-2.84)***	0.110 (3.19)***	-0.032 (-2.89)***
$Leverage_{i,t}$	0.012 (0.67)	-0.083 (-2.32)**	0.094 (2.89)***	-0.022 (-0.64)	-0.013 (-0.72)
Observations	82,116	82,116	82,116	82,116	82,116
R ²	42.21%	11.74%	8.37%	34.15%	66.24%

Table 3B: extended determinants

6.6 Table 4 and Table 5: tick sizes and analyst coverage

Read:

- Table 4 uses the 1997 tick-size reduction pilot to validate the noise component.
- The tick-size change is associated with increases in private information and declines in noise, especially among smaller pilot stocks.
- Table 5 uses exogenous analyst-coverage losses from broker mergers and closures.
- A coverage shock reduces public firm-specific information by about 1.47 percentage points and increases noise by about 0.71 percentage points.

Economic message: Microstructure improvements reduce noise; loss of analyst coverage reduces public information and increases noise. These external shocks validate the decomposition.

Table 4
Effects of the tick size changes on noise and information

Variable	Model 1 <i>PrivateInfo</i> _{<i>i,t</i>}	Model 2 <i>PrivateInfo</i> _{<i>i,t</i>}	Model 3 <i>Noise</i> _{<i>i,t</i>}	Model 4 <i>Noise</i> _{<i>i,t</i>}	Model 5 <i>NoiseShare</i> _{<i>i,t</i>}	Model 6 <i>NoiseShare</i> _{<i>i,t</i>}
Intercept	1.39 (64.83)***	3.32 (85.95)***	1.02 (66.15)***	4.74 (98.32)***	19.87 (53.31)***	48.66 (77.12)***
D_t^{POST}	0.15 (5.86)***	0.58 (11.12)***	0.25 (12.86)***	0.29 (5.31)***	-0.76 (-1.98)**	-11.63 (-17.01)***
$D_t^{POST} \times Q1_i$	0.43 (8.83)***		0.28 (5.58)***		-5.55 (-8.36)***	
$D_t^{POST} \times Q2_i$	0.16 (3.54)***		0.07 (1.93)*		-5.87 (-8.80)***	
$D_t^{POST} \times Q3_i$	0.08 (1.72)*		0.04 (1.20)		-3.56 (-5.14)***	
$Q1_i$	1.26 (36.53)***		2.48 (64.38)***		18.87 (31.69)***	
$Q2_i$	0.57 (17.03)***		0.92 (31.80)***		10.38 (16.55)***	
$Q3_i$	0.21 (6.58)***		0.38 (15.64)***		7.23 (11.49)***	
$D_t^{POST} \times \ln P_{i,t}$		-0.11 (-6.04)***		0.02 (0.80)		2.77 (11.80)***
$\ln P_{i,t}$		-0.56 (-40.45)***		-1.09 (-63.33)***		-7.77 (-34.67)***
Observations	15,440	15,440	15,440	15,440	15,440	15,440
R^2	20.3%	22.3%	45.8%	49.3%	11.0%	11.7%

This table reports the results from panel regressions of stock-year observations. Dependent variables are given in the column headings: *PrivateInfo*_{*i,t*} is the amount of stock-return variance attributed to private firm-specific information, *Noise*_{*i,t*} is the amount of stock-return variance attributed to noise, and *NoiseShare*_{*i,t*} is the share (percentage) of variance attributed to noise. The models examine the effects of tick size reductions from eighths of a dollar to sixteens of a dollar on June 24, 1997, using 2 years of data around the change (1996, 1998). D_t^{POST} is a dummy variable that takes the value of one after 1997 and zero before. $Q1_i$, $Q2_i$, and $Q3_i$ are dummy variables that indicate the price quartile to which the firm belongs (the highest-price quartile, $Q4_i$, is

Table 4: tick-size effects on noise and information

Table 5
Effect of analyst coverage on variance components

Variable	<i>MktInfoShare</i> _{<i>i,t</i>}	<i>PrivateInfoShare</i> _{<i>i,t</i>}	<i>PublicInfoShare</i> _{<i>i,t</i>}	<i>NoiseShare</i> _{<i>i,t</i>}
Intercept	7.52 (112.75)***	-3.89 (-66.40)***	-4.71 (-102.49)***	1.08 (34.07)***
<i>CoverageShock</i> _{<i>i,t</i>}	0.31 (0.43)	0.45 (0.70)	-1.47 (-2.96)***	0.71 (2.09)**
Observations	116,879	116,879	116,879	116,879
R^2	23.0%	7.6%	2.5%	5.4%
Fixed effects	Stock, Time	Stock, Time	Stock, Time	Stock, Time

This table reports the results from difference-in-differences regressions in which we examine the causal effect of an exogenous drop in analyst coverage (due to broker mergers and closures) on variance components. The dependent variables are shares of stock-return variance attributable to market-wide information (*MktInfoShare*_{*i,t*}), private

Table 5: analyst-coverage shock effects

6.7 Table 6: mutual fund fire sales and discretionary trading

Read:

- Discretionary mutual fund trading shocks increase market-wide information by 0.33 and private firm-specific information by 0.50.
- Discretionary trading reduces the noise share by about 0.98 percentage points.
- Expected, nondiscretionary trade shocks have much weaker effects.

Economic message: The decomposition distinguishes informed institutional trading from mechanical fire-sale pressure. Informed discretionary trading makes prices more informative and less noisy.

Table 6
Effect of mutual fund fire sales and discretionary trade shocks on variance components

Variable	<i>MktInfoShare</i> _{<i>i,t</i>}	<i>PrivateInfoShare</i> _{<i>i,t</i>}	<i>PublicInfoShare</i> _{<i>i,t</i>}	<i>NoiseShare</i> _{<i>i,t</i>}
Intercept	-0.50 (-7.49)***	1.93 (12.08)***	0.40 (2.85)***	-1.83 (-23.20)***
<i>DiscretionaryTradeShock</i> _{<i>i,t</i>}	0.33 (3.44)***	0.50 (4.71)***	0.16 (1.59)	-0.98 (-12.13)***
<i>ExpectedTradeShock</i> _{<i>i,t</i>}	-0.23 (-1.68)*	0.19 (1.39)	0.07 (0.54)	-0.03 (-0.32)
Observations	470,307	470,307	470,307	470,307
R^2	27.1%	10.0%	5.8%	7.1%
Fixed effects	Stock, Time	Stock, Time	Stock, Time	Stock, Time

This table reports the results from difference-in-differences regressions of stock-quarter observations in which we examine the causal effect of an exogenous shock to mutual fund trading of an underlying stock, driven by investor redemptions or inflows. The dependent variables are shares of stock return variance attributable to market-wide information (*MktInfoShare*_{*i,t*}), private firm-specific information (*PrivateInfoShare*_{*i,t*}), public firm-specific information (*PublicInfoShare*_{*i,t*}), and noise (*NoiseShare*_{*i,t*}). *DiscretionaryTradeShock*_{*i,t*} takes the value of one for stock-quarters that experience strong discretionary buying or selling by mutual funds in response to a redemption shock (a proxy for informed mutual fund trading, defined as the bottom and top deciles of mutual

Table 6: mutual fund trading shocks and variance components

7 Short summary

- The paper asks what drives stock-price movements: news, private information, public information, market-wide information, or noise.
- It builds a VAR-based Beveridge-Nelson permanent-transitory decomposition.
- Permanent price responses are interpreted as information; temporary deviations are interpreted as noise.

- The baseline decomposition assigns 31% of return variance to noise, 24% to private firm-specific information, 37% to public firm-specific information, and 8% to market-wide information.
- Noise declines dramatically after the mid-1990s, while firm-specific information rises.
- Traditional R^2 can mislead because it does not distinguish firm-specific information from firm-specific noise.
- Large stocks and institutionally held stocks have less noise and more private information.
- Tick-size reductions, analyst-coverage shocks, and mutual fund trading shocks validate the economic labels of the components.
- The paper provides a tool for studying market efficiency in a way that distinguishes information from noninformational price pressure.

8 Conclusion

8.1 Core conclusion

Brogaard, Nguyen, Putnins, and Wu show that stock price movements are driven by a mixture of market-wide information, private firm-specific information, public firm-specific information, and noise. Their main empirical conclusion is that U.S. stock prices have become less noisy and more information-driven since the mid-1990s.

8.2 Economic intuition

A stock return can be caused by a real change in efficient price or by a temporary deviation from efficient price. A buying shock may push price too high and later reverse; a public news shock may initially be underreacted to and then be incorporated gradually. Looking only at contemporaneous returns or R^2 cannot distinguish these cases. Dynamic impulse responses allow the paper to separate permanent information from transitory noise.

8.3 Incremental takeaway

The paper's most important incremental contribution is a measurement framework that reconciles market microstructure, asset pricing, and market-efficiency debates. It shows that price informativeness is not just about how much of a stock's return is explained by the market. It is about whether price changes are permanent information or temporary noise, and about whether the information comes from public news, private trading, or market-wide shocks.